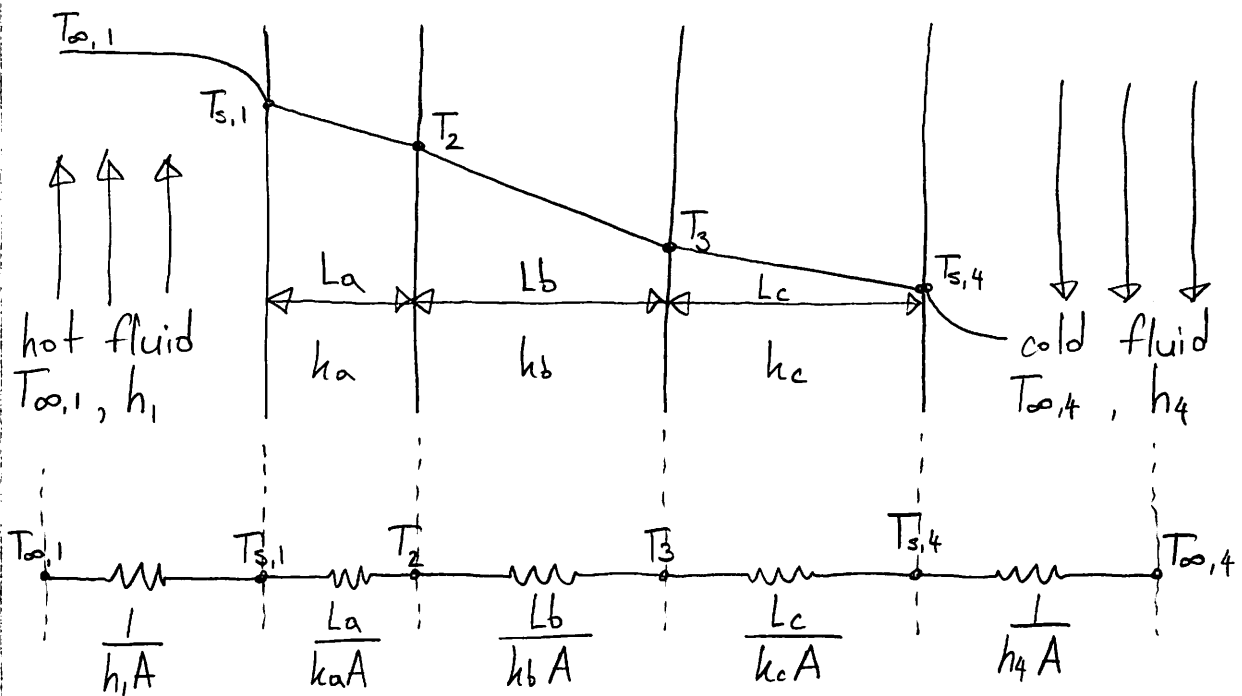


Composite Problems

Just like in circuits, we simply add up our thermal resistances to solve for heat transfer and temperature at any node.

For a composite wall:



So if we want the overall heat transfer, we know $T_{\infty,1}$ & $T_{\infty,4}$ and all material properties, we can sum the resistances in series:

$$Q_{TOT} = \frac{\Delta T_{TOT}}{R_{TOT}} = \frac{T_{\infty,1} - T_{\infty,4}}{\frac{1}{h_1 A} + \frac{L_a}{k_a A} + \frac{L_b}{k_b A} + \frac{L_c}{k_c A} + \frac{1}{h_4 A}}$$

Also, we can write:

$$Q_{TOT} = \frac{\Delta T}{R} = \frac{T_{\infty,1} - T_{s,1}}{\frac{1}{h_1 A}} = \frac{T_{s,1} - T_2}{\frac{L_a}{k_a A}} = \frac{T_2 - T_3}{\frac{L_b}{k_b A}} = \dots$$

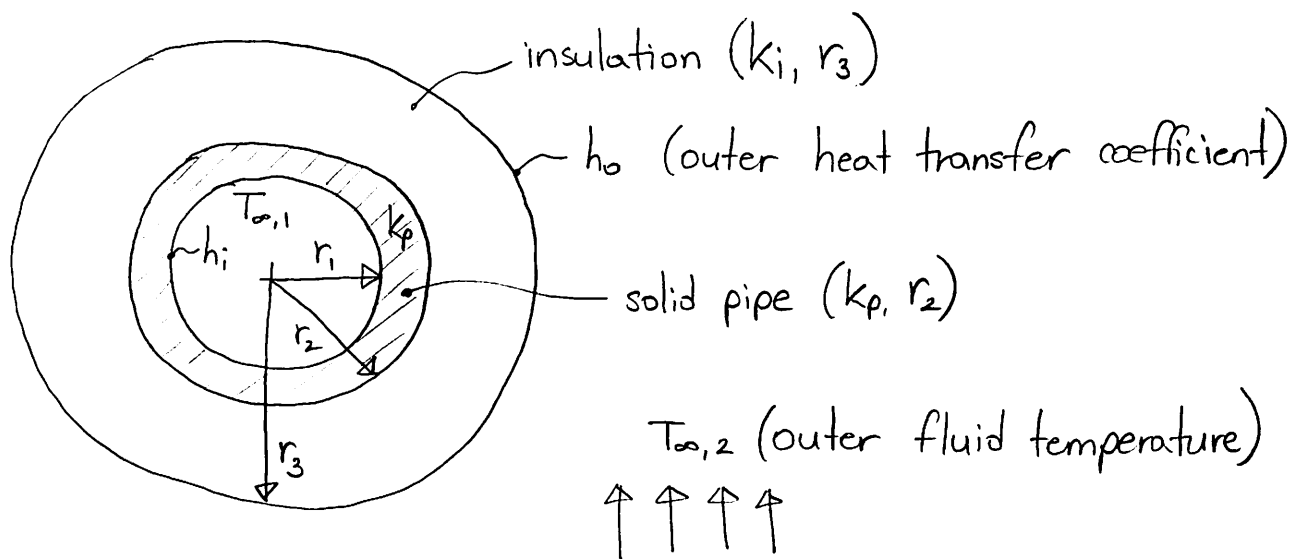
By doing the individual segment approach, we can then solve for intermediate temperatures $T_{s,1}, T_2, T_3, T_{s,4}$ which are typically unknowns in a design problem.

So in general, we can write: (for a wall with n layers)

$$Q = \frac{A (T_{\infty,1} - T_{\infty,n})}{\frac{1}{h_1} + \frac{1}{h_n} + \sum_{j=1}^n \frac{L_j}{k_j}}$$

$$R_{TOT} = \sum_{j=1}^n R_j$$

Critical Thickness of Insulation



What should r_3 be in order to minimize heat transfer from the fluid flowing within the tube to the outside fluid. Let's write out our total thermal resistance:

$$R_{TOT} = \underbrace{\frac{1}{2\pi r_1 h_i L}}_{\text{inner convection}} + \underbrace{\frac{\ln(r_2/r_1)}{2\pi k_p L}}_{\text{pipe conduction}} + \underbrace{\frac{\ln(r_3/r_2)}{2\pi k_i L}}_{\text{insulation conduction}} + \underbrace{\frac{1}{2\pi r_3 h_o L}}_{\text{outer convection}}$$

We can see from our equation that as we increase r_3 , the insulation conduction resistance increases, but the outer convection resistance decreases. So there must be an optimum!

We want to maximize R_{TOT} with respect to r_3 .

Remember, partial derivative

$$\frac{\partial R_{TOT}}{\partial r_3} = 0$$

$$\frac{\partial R_{TOT}}{\partial r_3} = \underbrace{\frac{2}{\partial r_3} \left(\frac{1}{2\pi r_i h_i L} \right)}_{=0 \text{ since } \neq f(r_3)} + \underbrace{\frac{2}{\partial r_3} \left(\frac{\ln(r_2/r_1)}{2\pi k_p L} \right)}_{=0 \text{ since } \neq f(r_3)} + \frac{2}{\partial r_3} \left(\frac{\ln(r_3/r_2)}{2\pi h_o L} \right) + \frac{2}{\partial r_3} \left(\frac{1}{2\pi h_o L} \right)$$

$$\frac{\partial R_{TOT}}{\partial r_3} = 0 = \frac{1}{2\pi h_o L} \frac{2}{\partial r_3} [\ln(r_3/r_2)] + \frac{1}{2\pi h_o L} \frac{2}{\partial r_3} \left[\frac{1}{r_3} \right]$$

$$\frac{1}{h_o} \frac{2}{\partial r_3} \ln\left(\frac{r_3}{r_2}\right) + \frac{1}{h_o} \frac{2}{\partial r_3} \left(\frac{1}{r_3} \right) = 0$$

$$\frac{1}{h_o} \cdot \frac{r_3}{r_3 \cdot r_2} + \frac{1}{h_o} \left(-\frac{1}{r_3^2} \right) = 0 \Rightarrow \text{Remember: } \frac{\partial}{\partial x} \ln(x) = \frac{1}{x}$$

$$\frac{1}{h_o} - \frac{1}{h_o r_3} = 0$$

$$\frac{\partial}{\partial x} \ln(ax) = \frac{1}{x} \cdot a$$

$$\boxed{r_{3,crit} = \frac{k_i}{h_o}} \Rightarrow \text{Critical thickness of insulation for a pipe.}$$

So is this a maximum or a minimum?

We can check with a second derivative.

$$\frac{\partial^2 R_{TOT}}{\partial r_3^2} = \left[\frac{2}{\partial r_3} \left(\frac{1}{h_o r_3} \right) - \frac{2}{\partial r_3} \left(\frac{1}{h_o r_3^2} \right) \right] \frac{1}{2\pi L}$$

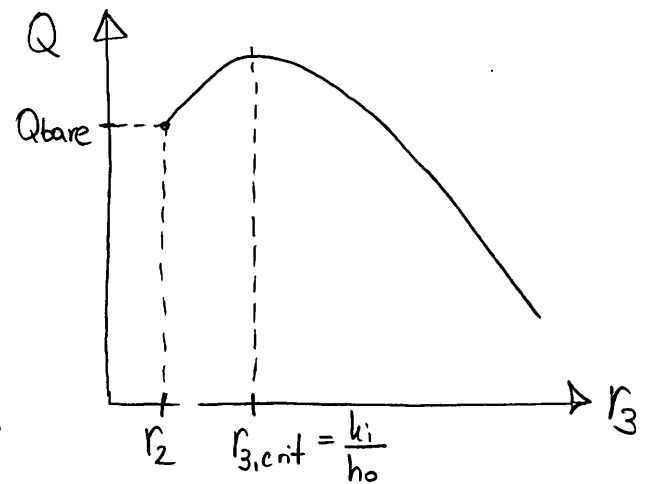
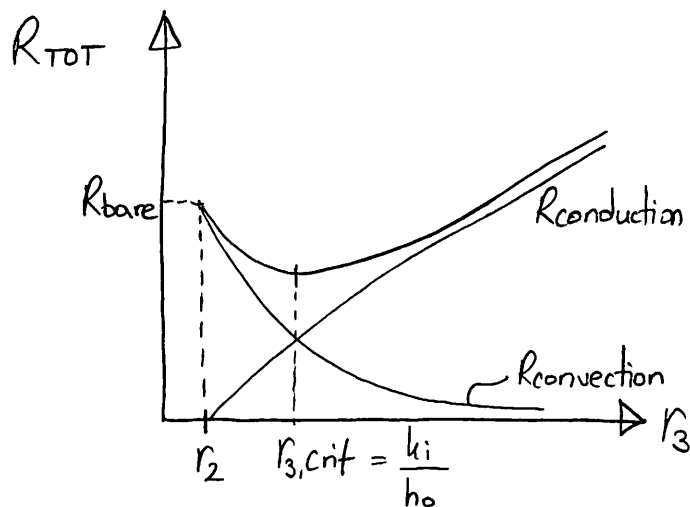
$$\frac{d^2 R_{TOT}}{dr_3^2} = \left(-\frac{1}{k_i r_3^2} + \frac{2}{h_o r_3^3} \right) \cdot \frac{1}{2\pi L} \quad (1)$$

Back substituting $r_{3,crit}$ into equation (1)

$$\left. \frac{d^2 R_{TOT}}{dr_3^2} \right|_{r_{3,crit}} = \left(-\frac{1}{k_i \left(\frac{k_i^2}{h_o^2} \right)} + \frac{2}{h_o \left(\frac{k_i^3}{h_o^3} \right)} \right) \frac{1}{2\pi L} = \left(-\frac{h_o^2}{k_i^3} + \frac{2h_o^2}{k_i^3} \right) \cdot \frac{1}{2\pi L}$$

$$\left. \frac{d^2 R_{TOT}}{dr_3^2} \right|_{r_{3,crit}} = + \frac{h_o^2}{2\pi k_i^3} > 0 \text{ (Always)}$$

So our calculated $r_{3,crit}$ is a global minimum.
No optimum thickness exists, only a critical insulation thickness.



This result tells us when it is ok to add insulation.
For example, for a hot water pipe, $r_2 = 10 \text{ cm}$

$$\left. \begin{array}{l} k_i = 0.1 \text{ W/m}\cdot\text{K} \\ h_o = 5 \text{ W/m}^2\cdot\text{K} \end{array} \right\} r_{3,crit} = \frac{0.1 \text{ W/m}\cdot\text{K}}{5 \text{ W/m}^2\cdot\text{K}} = 0.02 \text{ m} = 2 \text{ cm}$$

Here $r_{3,crit} < r_2 \Rightarrow$ means it is OK to insulate!